Zhengzhong Ricky You

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Research Interests

- Exact algorithms for large-scale integer and combinatorial optimization, with an emphasis on branch-price-and-cut, decomposition, and extended formulations.
- Learning-augmented optimization, decision-focused learning, and reinforcement learning for combinatorial optimization, together with interfaces between operations research and machine learning.
- Applications of large language models to healthcare operations, including routing, scheduling, and resource allocation under uncertainty and fairness constraints.

Education

- **University of Florida**, M.S. in Operations Research **2025**
- **Nankai University**, B.S. in Chemistry **2021**

Publications

1. Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly.
Zhengzhong Ricky You, Yu Yang, Xinshang Wang, Wotao Yin. *Operations Research*, 2026.
2. RouteOpt: An Open-Source Modular Exact Solver for Vehicle Routing Problems.
Zhengzhong Ricky You, Yu Yang. *INFORMS Journal on Computing*, 2026.
3. Equity-Enforced Path Recommendation in Congested Transit Systems During Disruptions.
Baichuan Mo, **Zhengzhong Ricky You**, Ruimin Li. Major revision at *Transportation Science*.
4. Bulk Service Queueing for Transit Resilience under Short Random Service Suspensions.
Baichuan Mo[†], **Zhengzhong Ricky You**[†], Li Jin, Zuo-Jun Max Shen, Haris N. Koutsopoulos, Jinhua Zhao. Reject and resubmit at *Transportation Science*.
5. Learned Pairwise Deep Dual-Optimal Inequalities for Stabilizing Column Generation.
Zhengzhong Ricky You, Bo Tang, Haoran Liu, Baichuan Mo. Under review at *INFORMS Journal on Computing*.
6. PrecPack: An Efficient Open-Source Exact Solver for Bin Packing with Generalized Precedence Constraints.
Sunkanghong Wang, **Zhengzhong Ricky You**, Roberto Baldacci, Baichuan Mo, Hu Qin, Lijun Wei, Zhou Xu. Under review at *INFORMS Journal on Computing*.
7. KKT Atlas: Learning Low-Dimensional Local Scenario Models for Contextual Stochastic Optimization.
Zhengzhong Ricky You, Baichuan Mo, Jinhua Zhao, Zuo-Jun Max Shen. Working paper. To be submitted to *Operations Research*.
8. [‡] Debiasing Optimal Values in Stochastic Programming by Averaging Solutions.
Henry Lam, Huajie Qian, Kaizhao Sun, **Zhengzhong Ricky You**. Working paper. To be submitted to *Operations Research*.
9. Driver Surge Maps for Supply Rebalancing with Downstream Matching.
Yida Xu[†], Baichuan Mo[†], **Zhengzhong Ricky You**[†], Yuan Zhou, Hao Yi Ong, Zuo-Jun Max Shen. Working paper. To be submitted to *Manufacturing & Service Operations Management*.
10. TorchDCM: A Unified PyTorch-Native Package for Discrete Choice Modeling.
Baichuan Mo, **Zhengzhong Ricky You**[*], Xiqun Michael Chen, Ruimin Li.
11. A Dual-Guided Column-Generation-Based Matheuristic for Integrated Railroad Blocking.
Yida Xu[†], **Zhengzhong Ricky You**[†], Baichuan Mo. Working paper. To be submitted to *Transportation Science*. Finalist, 2026 INFORMS RAS Problem Solving Competition.
12. Log-Driven Semantic Tuning for Mixed Integer Programming Solvers.
Zhengzhong Ricky You, Kaizhao Sun, Zeping Min, Wotao Yin, Xinshang Wang. Working paper.
13. Draft-and-Audit Reinforcement Learning for Optimization Modeling.
Zeping Min, Weihang Xu, **Zhengzhong Ricky You**, Wotao Yin, Xinshang Wang. *ICML 2026*.
14. Towards Routing Foundation Models for Commercial Vehicles.
Bo Tang, Yunfei Ma, **Zhengzhong Ricky You**. *UMFM 2026*, ACM SIGSPATIAL.

[*] Corresponding author. [†] Co-first authors; contributed equally. [‡] Authors listed alphabetically by last name.

Research Experience

- **TikTok Inc.** **2026**
  Machine Learning Engineer Intern, E-commerce Supply Chain & Logistics *San Jose, CA*
  - Researched heuristic algorithms for inventory control in e-commerce supply chains, under the mentorship of Baichuan Mo, Yikang Hua, and Zhou Ye.
  - Developed an LLM-based agent to analyze algorithm outputs and investigate the reasons behind observed results, under the mentorship of Yufei Ruan.
- **MIT-UF-NEU Joint Summer Research Camp** **2026**
  Selected Participant, MIT JTL-Transit Lab program *Remote*
  - Developed **KKT Atlas** for contextual stochastic optimization under the mentorship of Baichuan Mo. Combined local context partitioning, gradient-informed dimension reduction, and joint learning of compact scenario supports and context-dependent probabilities.
  - Analyzed sufficient conditions under which a d-dimensional local uncertainty representation admits $d+1$ shared scenarios. Derived decision-error bounds and quantified the tradeoff between finer localization and estimation error from smaller local samples.
  - Evaluated the method on capacity allocation, portfolio optimization, and network reservation problems. Used 0.8–3.6% of the reference support size and reduced deployment time by 92–98% relative to the full-support comparator, with the lowest or nearly lowest mean risk among the evaluated baselines.
- **University of Kentucky** **2026**
  Visiting Scholar, Department of Statistics *Lexington, KY*
  - Conducted research on online learning algorithms during a visit hosted by Chenglong Ye.
- **DAMO Academy, Alibaba Group** **2025**
  Research Intern *Bellevue, WA*
  - Studied solver parameter tuning with large language models under the mentorship of Xinshang Wang and Wotao Yin. The resulting configurations improved average benchmark performance relative to solver defaults, whereas tested black-box baselines generally did not.

Research Software

- **PrecPack** **2026**
  Co-developer, C++ *Open source*
  - Co-developed an exact solver for bin packing with generalized precedence constraints, combining branch-bound-and-remember search, independently checked solutions, and reproducible batch workflows.
- **TorchDCM** **2026**
  Co-developer, Python *Open source*
  - Co-developed a unified PyTorch-native package for discrete choice modeling. The package implements vectorized likelihoods and automatic differentiation across multiple logit model families and provides standard econometric outputs.
- **RouteOpt** **2021**
  Lead Developer, 20k+ LOC in C++ *Open source*
  - Designed and implemented an open-source exact solver for the CVRP and CVRPTW. The solver uses modular branching, cutting-plane, and variable-reduction components, learning-augmented branching, and reproducible experiment scripts.
  - On benchmark instances, RouteOpt nearly doubled the speed of VRPSolver and proved optimality for four previously open instances; one required approximately 1.5 years of single-core time.

Selected Relevant Courses

- **ISE Ph.D. level (GPA: 4.00)**: Discrete Optimization; Linear Programming and Network Optimization; Stochastic Modeling & Analysis; Advanced Topics in Continuous Optimization.
- **Mathematics Ph.D. level (GPA: 4.00)**: Analysis I; Numerical Linear Algebra; Numerical Analysis; Numerical Optimization.
- **CS Ph.D. level (GPA: 4.00)**: Machine Learning.

Conference & Workshop Presentations

- **Finalist presentation (upcoming)**, INFORMS Railway Applications Section (RAS) Problem Solving Competition, INFORMS Annual Meeting, San Francisco, November 1–4 **2026**
- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Transportation Science and Logistics Conference, MIT Sloan **2026**
- *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*, invited talk, UK AI and ML Symposium and Nontechnical Workshop **2025**
- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Annual Meeting **2025**
- *Accelerating Dynamic Programming via Dual Selection for Column-Generation Frameworks*, INFORMS Optimization Society Conference **2024**
- *RouteOpt: A Scalable Advanced Optimization Tool for VRPs*, INFORMS Annual Meeting **2024**
- *Two-Stage Learning to Branch in Branch-Price-and-Cut for Exact VRP*, INFORMS Annual Meeting **2023**
- *Learning to Branch with Column Generation*, INFORMS Annual Meeting **2022**

Awards & Honors

- **Finalist**, INFORMS Railway Applications Section (RAS) Problem Solving Competition; one of five finalist teams selected from 26 final-report submissions **2026**
- Harold D. Haldeman, Jr. Graduate Fellowship, University of Florida **2024**
- INFORMS Optimization Society Conference Travel Support **2024**
- Academic Excellence Scholarship, Nankai University **2020**
- Innovation and Entrepreneurship Scholarship, Nankai University **2019**
- First Prize, Tianjin Theoretical Chemistry Competition **2019**
- First Prize, Tianjin Mathematics Competition **2018**

Teaching & Mentoring

- **Sole Instructor**, ESI 3327C Matrix & Numerical Methods in Systems Engineering, University of Florida; taught linear algebra, numerical methods, and introductory optimization to 52 undergraduates **2024**
- **Outreach Mentor**, UF Student Science Training Program; Ken Cheng, Columbia University (computer science), and Shrey Gupta, University of Oxford (mathematics)

Professional Service

Conference reviewer for the *International Conference on Learning Representations (ICLR)*. Journal referee for *INFORMS Journal on Computing*, *INFORMS Journal on Optimization*, *Transportation Science*, *Transportation Research Part B*, *IISE Transactions*, *Computers & Operations Research*, and *OR Spectrum*.

Skills

Methods: branch-price-and-cut, column generation, dual stabilization, cutting planes, learning-augmented optimization, stochastic and robust optimization. **Programming**: C++, Python including PyTorch and reinforcement learning libraries, MATLAB, Shell, and Git. **Software**: PrecPack, RouteOpt, TorchDCM, Gurobi, CPLEX, SCIP, and reproducible experiment pipelines.